

OS PRACTICAL 3

Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution. Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

created a C program using the fork() system call. The fork() function creates a new child process from the existing parent process.

```
samhi@Samhita:~/practical_3$ nano fork.c
samhi@Samhita:~/practical_3$ gcc fork.c
samhi@Samhita:~/practical_3$ ./a.out
this line is repeating twice because of fork()
this line is repeating twice because of fork()
samhi@Samhita:~/practical_3$ |
```

created the child process displayed the PID and PPID of both the parent and child processes to identify them separately.

```
samhi@Samhita:~/practical_3$ nano fork1.c
samhi@Samhita:~/practical_3$ gcc fork1.c
samhi@Samhita:~/practical_3$ ./a.out
Parent Process
Parent Process ID is 891
Parent's Parent Process ID is 336
Child Process
Child Process ID is 892
Parent Process ID is 335
samhi@Samhita:~/practical_3$ |
```

During the execution of the program, the ps command was used to view the processes currently running in the terminal. It displayed the process IDs and the corresponding commands.

```
samhi@Samhita:~/practical_3$ gcc fork1.c -o fork1
samhi@Samhita:~/practical_3$ ./fork1
Parent Process
Parent Process ID is 1791
Parent's Parent Process ID is 336
Child Process
Child Process ID is 1792
Parent Process ID is 1791
samhi@Samhita:~/practical_3$ ps
  PID TTY          TIME CMD
  336 pts/0        00:00:00 bash
 1792 pts/0        00:01:12 fork1
 1819 pts/0        00:00:00 ps
samhi@Samhita:~/practical_3$ |
```

Used the top command to continuously monitor the processes while they were executing. It provided a real-time view of the processes and their activity. This helped in observing how the process status changes during execution.

```
samhi@Samhita:~/practical_3$ ps
  PID TTY          TIME CMD
  336 pts/0    00:00:00 bash
 1792 pts/0    00:01:12 fork1
 1819 pts/0    00:00:00 ps
samhi@Samhita:~/practical_3$ top
top - 15:30:43 up 1:15, 1 user, load average: 0.93, 0.52, 0.29
Tasks: 28 total, 2 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 6.2 us, 0.0 sy, 0.0 ni, 93.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7755.7 total, 7081.9 free, 566.2 used, 259.6 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7189.5 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 1792 samhi     20   0   2768   1068   968 R 100.0   0.0   2:37.19  fork1
    1 root      20   0   24552 15488 11532 S   0.0   0.2   0:00.61  systemd
    2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.00  init-systemd(Ub
    6 root      20   0    3180   2044    1940 S   0.0   0.0   0:00.00  init
   46 root      19  -1   50396 16824 15552 S   0.0   0.2   0:00.25  systemd-journal
   78 systemd+  20   0   22416 14496 12004 S   0.0   0.2   0:00.08  systemd-resolve
   85 root      20   0   35412 12356   9224 S   0.0   0.2   0:01.53  systemd-udev
  151 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.03  chronyd-starter
  155 root      20   0    4472   3096   2852 S   0.0   0.0   0:00.01  cron
  156 message+  20   0    8796   5380   4692 S   0.0   0.1   0:00.05  dbus-daemon
  163 root      20   0   44096 29864 14724 S   0.0   0.4   0:00.18  networkd-dispat
  170 root      20   0   18440   9448   8200 S   0.0   0.1   0:00.09  systemd-logind
  192 syslog    20   0  220548   5780   4496 S   0.0   0.1   0:00.07  rsyslogd
  236 _chrony    20   0   23588 11004   7088 S   0.0   0.1   0:00.13  chronyd
  239 root      20   0  123460 32780 18172 S   0.0   0.4   0:00.13  unattended-upgr
  251 _chrony    20   0   12096   2460   1820 S   0.0   0.0   0:00.00  chronyd
  294 root      20   0    5492   2692   2508 S   0.0   0.0   0:00.00  agetty
  334 root      20   0    3184   1112    980 S   0.0   0.0   0:00.00  SessionLeader
  335 root      20   0    3200   1252   1108 S   0.0   0.0   0:00.24  Relay(336)
  336 samhi     20   0    6268   5556   3864 S   0.0   0.1   0:00.30  bash
  337 root      20   0    8644   5604   4764 S   0.0   0.1   0:00.01  login
  380 samhi     20   0   22340 13600 10988 S   0.0   0.2   0:00.10  systemd
  382 samhi     20   0   22920 4092  2088 S   0.0   0.1   0:00.00  (sd-pam)
  408 samhi     20   0    6136   5468   3864 S   0.0   0.1   0:00.02  bash
  950 root      20   0    3184   1120    980 S   0.0   0.0   0:00.00  SessionLeader
  951 root      20   0    3200   1132    980 S   0.0   0.0   0:00.01  Relay(956)
  956 samhi     20   0    6268   5552   3872 S   0.0   0.1   0:00.10  bash
```

Used the /proc filesystem to inspect the process with PID 1792, which was the fork1 process identified using the ps command. The /proc/1792 directory contains different files and directories that provide information about the process, such as its status, memory, scheduling, file descriptors and other process-related details. This helped me observe how Linux maintains information about an active process.

```
samhi@Samhita:~/practical_3$ ls /proc/1792
arch_status  cmdline      environ      io            loginuid      mounts      oom_adj      personality  sessionid    stat          timens_offsets
attr         comm         exe          ksm_merging_pages  map_files     mountstats  oom_score    projid_map   setgroups    statm         timers
auxv         coredump_filter fd           ksm_stat      maps          net         oom_score_adj root         smaps        status        timerslack_ns
cgroup       cpuset       fdinfo       latency       mem           ns          pagemap      sched        smaps_rollup syscall       uid_map
clear_refs   cwd          gid_map      limits        mountinfo     numa_maps   patch_state  schedstat    stack        task          wchan
samhi@Samhita:~/practical_3$
```